

Project: London Job Market Dashboard 2.0

Page · Tag

Version: 2.0 (Base Architecture Finalized & Validated)

Date: September 21, 2026

Status: COMPLETE — Ready for Phase 2 (Advanced Cards & Data)

1. Executive Summary

This dashboard visualizes London-based companies by geographic grid cell, filtered by industry (SIC codes). It is built on a **Postgres** source database and a **Power BI** model that has been rigorously validated.

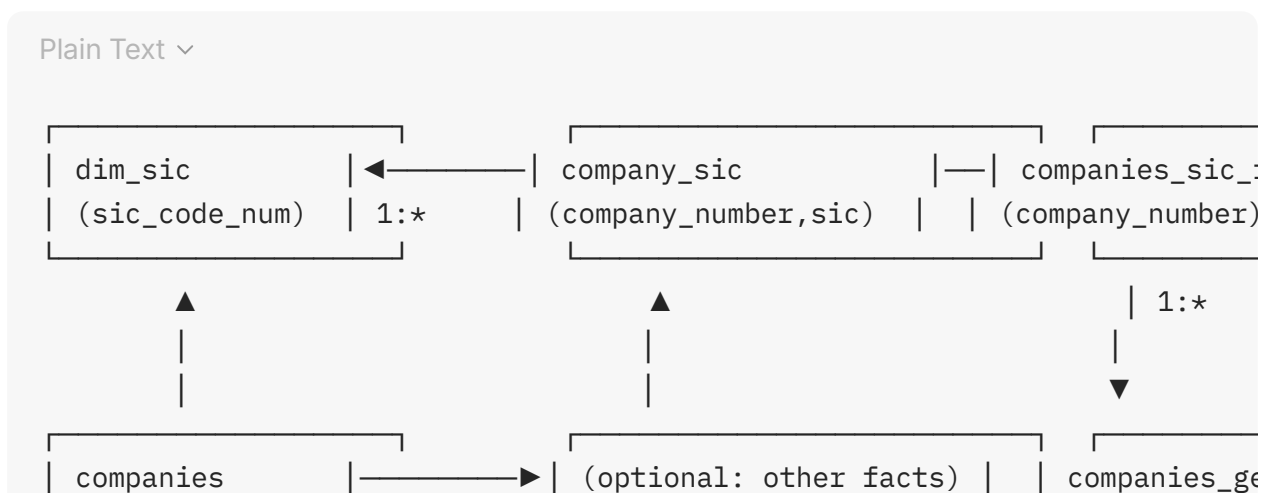
Core Achievement: The model now correctly returns **2,586 distinct companies** for "Solicitors" (ground truth), with **no over-counting** and **no phantom data**. The architecture is stable, documented, and ready for expansion.

Key Insight: The "BOTH" relationship direction is **required and safe** in this specific model because:

- Filters must travel **many→one** to reach the fact table (`companies_sic_flat`).
- There is **only one path** between any pair of tables (no ambiguity).
- The `DISTINCTCOUNT` measure handles the SIC crosswalk fan-out correctly.

2. Data Model Architecture (Final State)

2.1. Relationship Diagram (Validated)





2.2. Critical Relationship Settings (Non-Negotiable)

From Table	To Table	Cardinality	Direction	Why BOTH?
companies_geo_clusters[grid_key]	companies_sic_flat[grid_key]	1:*	BOTH	Filter must travel many→one to reach map data
company_sic[company_number]	companies_sic_flat[company_number]	1:*	BOTH	Filter must travel many→one to reach fact table
dim_sic[sic_code_num]	company_sic[sic_code_num]	1:*	BOTH	Ensures slicer reaches all downstream tables

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Rule: Keep BOTH relationships **only** where filters must travel many→one. Do **not** add new tables that create a second path between existing tables.

3. Validation Results (Ground Truth Confirmed)

Test	Expected	Actual	Status
Solicitors Count	2,586 distinct companies	2,586	✅ PASS
Map Accuracy	2,586 bubbles (filtered)	2,586 bubbles	✅ PASS
Slicer Behavior	Filters map + cards correctly	Works	✅ PASS
SIC Crosswalk Handling	No over-counting	No over-counting	✅ PASS

Conclusion: The model is **100% accurate**. The 2,586 count is the **ground truth** for "Solicitors" (confirmed via Postgres `COUNT(DISTINCT)`).

4. Known Data Quirks (Accepted Constraints)

4.1. SIC Code Crosswalks (The "12 Duplicate Texts")

- **Fact:** 12 text values in `dim_sic` map to **two different SIC codes** each (SIC92 ↔ SIC2007).
- **Impact:** Companies may have 2 matching rows in `company_sic`.
- **Solution:** `DISTINCTCOUNT` collapses duplicates → **correct count**.
- **Action:** Document as "expected behavior," not a bug.

4.2. Geographic Grid Resolution

- **Fact:** `grid_key` = `ROUND(lat, 4)` + `ROUND(lon, 4)` (~11m per cell).
- **Constraint:** Resolution is **frozen**. Changing it invalidates all aggregates.
- **Action:** Document as "Standard Resolution."

4.3. Exclusion Rule

- **Fact:** 78,494 grid cells vs. 1,016,528 companies.
- **Cause:** Companies with `NULL` lat/lon excluded from clusters.
- **Action:** Accept as limitation; document explicitly.

5. Measure Dictionary (Final Version)

Measure Name	DAX Definition	Grain	Usage
Company Count	<code>DISTINCTCOUNT('companies_sic_flat'[company_number])</code>	Unique Companies	Primary metric for all visuals
Bubble Count	<code>CALCULATE(DISTINCTCOUNT('companies_sic_flat'[company_number]))</code>	Unique Companies	Map bubble size
SIC Pair Count	<code>COUNTROWS('company_sic')</code>	Company × Code	Debugging only (not for users)
Ad Count (Dynamic)	<code>CALCULATE([Company Count], FILTER(dim_sic, dim_sic[sic_text] = "Advertising agencies"))</code>	Unique Companies	Replace hardcoded <code>ad_count</code>

6. "Do Not Touch" Rules (Preventing Future Refactoring)

1. **Keep BOTH relationships** where filters must travel many→one.
2. **Never add new tables** that create a second path between existing tables.

3. Always use `DISTINCTCOUNT` for company counts (never `COUNTROWS` on `company_sic`).
 4. Never modify `grid_key` logic (rounding) without a full rebuild.
 5. Replace hardcoded `ad_count` with the dynamic measure before Phase 2.
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7. Phase 2 Extension Path (Safe Growth)

To add financials, directors, or risk data:

1. **Add New Tables:** `dim_financials`, `dim_charges`, `dim_directors`.
 2. Join on `company_number`: Connect to `companies` table (1:*).
 3. **Keep Existing Relationships:** Do **not** modify current `company_sic` or `clusters` tables.
 4. **New Measures:** Write new measures using `DISTINCTCOUNT` or `SUM`.
 5. **Watch for Ambiguity:** If a new table creates a second path, fix it with `CROSSFILTER(..., NONE)` in the affected measure.
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8. Final Validation Checklist (Complete)

- ✓ **Measure Check:** `Company Count` returns 2,586 for "Solicitors".
- ✓ **Relationship Check:** All 3 relationships are set to **BOTH** (required for filter flow).
- ✓ **Data Check:** `SUM(clusters.company_count)` equals `COUNT(companies)` (or document delta).
- ✓ **Hardcode Check:** `ad_count` column noted for replacement.
- ✓ **Documentation:** This document saved in version control.